

M90 PRO

LITHIUM IRON PHOSPHATE BATTERY



320Ah
51.2V

MOTOMA®

Power into the Future

Features

- Grade A+ Cells**
Manufactured using the premium quality, fresh LiFePO4 raw materials for top-tier performance.
- High Cycle Efficiency**
Achieve 8000 cycles at 80% DoD, ensuring a cost-effective ownership experience.
- Advanced Touch Screen**
Multi-language colorful touch screen for clear monitoring of battery parameters, with easy control over settings.
- Wi-Fi Monitoring**
Enables remote battery monitoring, full control of settings, and easy report generation through a wireless connection.
- Extended Durability**
Enjoy a longer lifespan with our low-maintenance, stable chemistry batteries.
- Integrated Safety Measures**
Enhanced with a built-in Battery Management System (BMS) to prevent misuse.
- Heat Resilience Engineered for optimal**
Engineered for optimal performance, even in extreme temperatures up to +60°C.
- Superior Safety**
Designed with advanced safety features to guarantee secure usage and storage.

Smart BMS Compatible with multiple inverter brands.



Introduction

MOTOMA M series LFP battery designed with 15 years or more service life for general purpose, which designed with advanced technology, built in intelligent BMS for more safe, with SOC design and communication protocol to match up with different brand inverters. Also motoma can supply customized upper computer software for BMS communication via RS232 to set parameters or read monitoring data.

Applications

- ◆ Power Generation System (Solar and Wind Power System, etc.)
- ◆ Household Energy Storage System
- ◆ Auto Control System & ATM Machine
- ◆ Electronic Apparatus and Equipment
- ◆ Emergency Light & Emergency Backup
- ◆ Power Supply & Alarm / Security System
- ◆ Communication Power & DC Power
- ◆ Electric Power System (EPS)
- ◆ Uninterruptible Power System (UPS)

Certificates



Technical Parameter

Battery Model	M90 PRO
Nominal Voltage	51.2V
Capacity	320Ah
Nominal Capacity	16.384kWh
Operating Voltage Range	43.2V~58.4V
Maximum Charging Current	200A
Maximum Discharging Current	200A
Charging Temperature Range	0°C~45°C
Discharging Temperature Range	-20°C~60°C
IP Level	IP20
Cell Cycle Life (0.5C/0.5C, RT 25°C)	8000 time
Communication Mode	RS485 / CAN / RS232
Battery Dimensions (LxWxH)	990*505*200 mm
Battery Weight (NW)	126.6 kg

Charge Performance

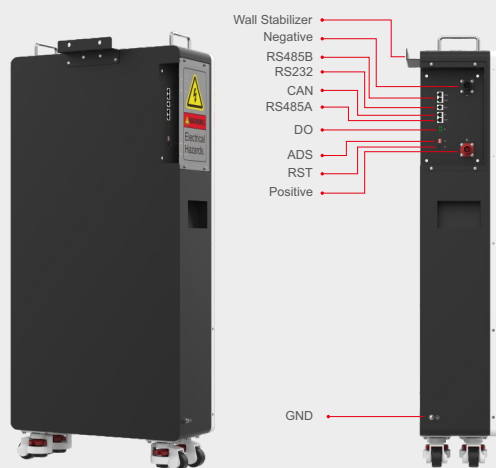
Recommended Charge Current	30A
Maximum Charge Current	200A
Peak Charge Current	210A (1s)
Recommended Charge Voltage	57.6V
BMS Charge Cut-Off Voltage	58.4V (3.65V/Cell)
Reconnect Voltage	54.08V (3.38V/Cell)
Balancing Voltage	54.4V (3.4V/Cell)
Balancing Current	40mA

Discharge Performance

Recommended Discharge Current	30A
Maximum Continuous Discharge Current	200A
Peak Discharge Current	210A (10s)
BMS Discharge Cut-Off Current	250A (500ms)
Recommended Low Voltage Disconnect	48V (3.0V/Cell)
BMS Discharge Cut-Off Voltage	43.2V (2.7V/Cell)
Reconnect Voltage	49.6V (3.1V/Cell)
Short Circuit Protection	250~500us

Packaging

Packaging Type	1 Battery / UN Wooden Box
Package Dimension (LxWxH)	1240x605x370 mm
Package Weight (GW)	156.6 KG



Grade A⁺ Cells

8000

≥ 8000 Cycles



Support Wi-Fi
monitoring

SOC

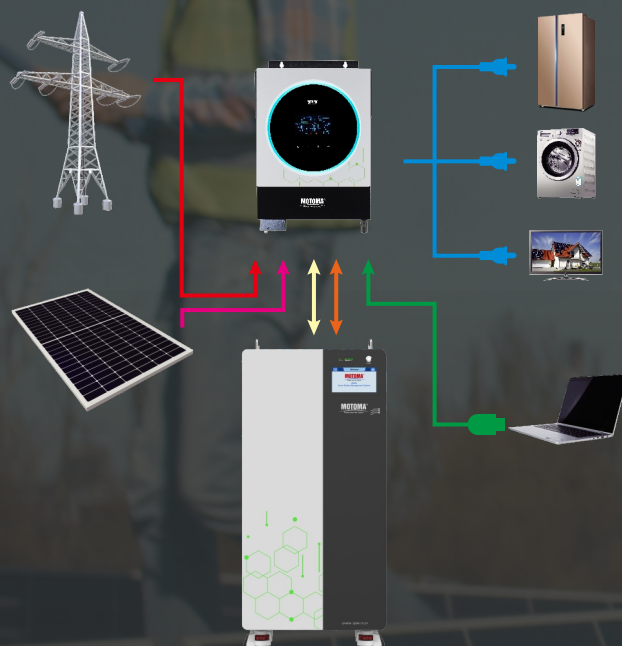
With State of
Charge design



Support high discharge
current up to 150A



Max. 15 pcs batteries
in parallel



- AC Input
- PV Input
- Power Cable
- Data Cable
- Com. Cable
- AC Output



Technical Support for Lifetime



7 Years Warranty